

Title: - Predicting the Habitability of Exoplanets Using Machine Learning

Link of website: - <https://inspiring-haupia-ea27aa.netlify.app/>

Snapshots of website-



This screenshot displays the 'Habitability Prediction' form on the ExoHabit website. The form is divided into two main sections: 'Planetary Parameters' and 'Stellar Parameters'. The 'Planetary Parameters' section includes input fields for Planet Radius (Earth radii), Planet Mass (Earth masses), Orbital Period (days), Semi-Major Axis (AU), Equilibrium Temp (K), and Planet Density (g/cm³). The 'Stellar Parameters' section includes input fields for Stellar Temp (K), Stellar Luminosity (solar), Stellar Metallicity (Fe/H), and Star Type. Below these fields are two buttons: 'Fill Example (Habitable)' and 'Fill Example (Not Habitable)'. At the bottom of the form, there are two buttons: 'Predict Habitability' and 'Reset'.

Planetary Parameters		Stellar Parameters	
Planet Radius (Earth radii)	Planet Mass (Earth masses)	Stellar Temp (K)	Stellar Luminosity (solar)
2.759	6.63	5320	0.519
Orbital Period (days)	Semi-Major Axis (AU)	Stellar Metallicity (Fe/H)	Star Type
9.15059	0.0814	-0.02	Type K — Orange dwarf
Equilibrium Temp (K)	Planet Density (g/cm³)		
828	1.73		



Potentially Habitable

The exoplanet is predicted to be potentially habitable with a confidence score of 100.00%.

Habitability Score

100.00%

PREDICTION

Potentially Habitable

CLASS

1

CONFIDENCE

100.00%

RAW SCORE

0.999990



Habitability Rankings

Top exoplanets ranked by their predicted habitability scores

Show top:

All

Load Rankings



Habitability Rankings

Top exoplanets ranked by their predicted habitability scores

Show top:

10

Load Rankings

RANK	SCORE	RADIUS	MASS	ORBITAL PERIOD	EQ. TEMP (K)	DENSITY	STELLAR TEMP	STATUS
#1	99.999%	2.759	6.63	9.15	828	1.73	5320	Habitable
#2	99.999%	2.75	10.60	9.43	794	2.90	5230	Habitable
#3	99.999%	3.12	11.84	16.61	499	2.14	4227	Habitable
#4	99.999%	2.7438	16.00	22.41	529	4.24	5165	Habitable
#5	99.999%	2.85	12.40	7.37	694	2.90	4491	Habitable
#6	99.999%	2.363	10.30	22.41	525	4.30	4796	Habitable
#7	99.999%	2.6615	10.60	9.61	759	3.05	5457	Habitable
#8	99.999%	2.592	15.90	13.86	638	5.02	5010	Habitable
#9	99.999%	2.62	23.50	22.41	469	6.82	4869	Habitable
#10	99.999%	2.41	14.90	22.41	470	5.80	5368	Habitable